

EE 66
Fall 2026

Signals, Dynamics, and Information

Homework 2

This homework is due Friday, September 11, 2026, at 23:59.

Submission Format

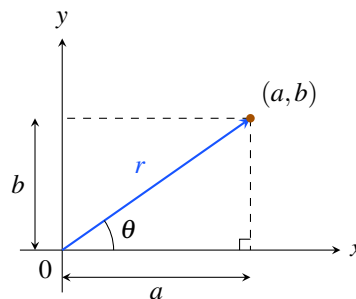
Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned).

On occasion, a problem set contains one more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

Submit the file to the appropriate assignment on Gradescope.

1. Complex Subspaces

Recall (from your physics classes, or vector calculus), we can describe a 2D vector as a point in 2D space. Sometimes we describe that point in “cartesian” coordinates, breaking it down into components along the orthogonal x and y axes. But other times, it is more convenient to describe the same vector in “polar” coordinates; here, we use its length r , and the angle θ formed between it and the positive x axis. Convince yourself that both formats can represent any point in 2D space equally well!



We use a *very similar* “change of coordinates” to represent complex numbers in two different ways:

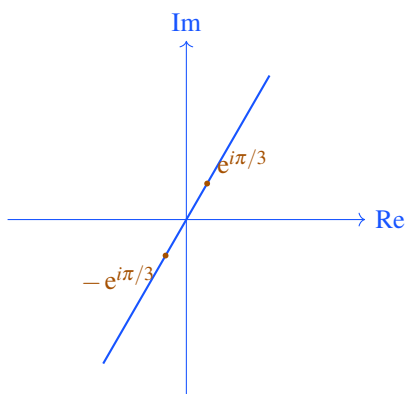
$$z = a + bi \iff z = r e^{i\theta}$$

In this problem, we associate with each complex number $z = a + bi$ a real vector $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$. In each part, you will consider a particular set of complex numbers. Determine if the associated set of vectors in $\mathbb{R}^2$ is a subspace. Justify your conclusion; and if it is a subspace, provide a basis for it.

Hint: try drawing each set of points in the complex plane, and justify your conclusion graphically.

(a) $S = \{v \mid v = r e^{i\pi/3}, r \in \mathbb{R}\}.$

Solution: Yes, S is a subspace. We can draw it like this (the points in orange correspond to $r = \pm 1$):



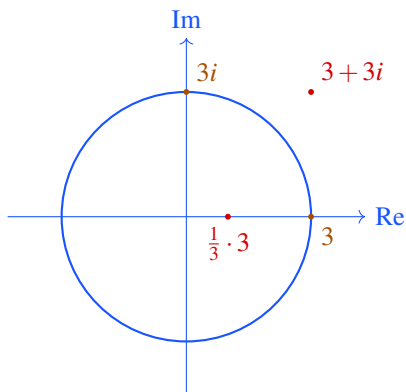
We see a line that *passes through the origin*, which clearly contains 0. Adding two points on this line, or scaling any point on this line, gives another point on the same line. To find a basis, we can use the fact that $e^{i\pi/3} = \cos(\pi/3) + i\sin(\pi/3) = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, so the set can be written

$$\left\{ r \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix} \mid r \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \right\}.$$

Hence, a valid basis is $\left\{ \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \right\}$.

- (b) $T = \{v \mid v = 3e^{i\theta}, \theta \in [0, 2\pi)\}$.

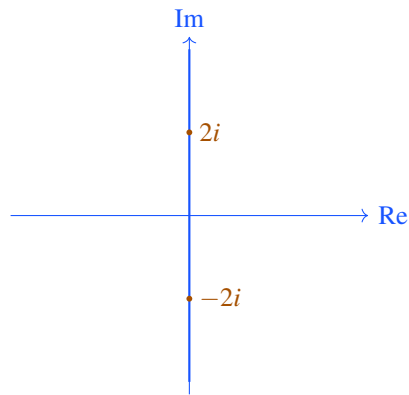
Solution: No, T is not a subspace. We can draw it like this:



This is a circle of radius 3 centered at the origin, so 0 is *not* in the set. Further, adding two points on this circle, or scaling any point on this circle, generally yields a point *outside* the circle. For example, 3 and $3i$ each lie in the the subspace (shown in red). However, neither of $\frac{1}{3} \cdot 3 = 1$ or $3 + 3i$ (shown in orange) lie in the subspace.

- (c) $U = \{v \mid v = bi, b \in \mathbb{R}\}$.

Solution: Yes, U is a subspace. We can draw it like this:



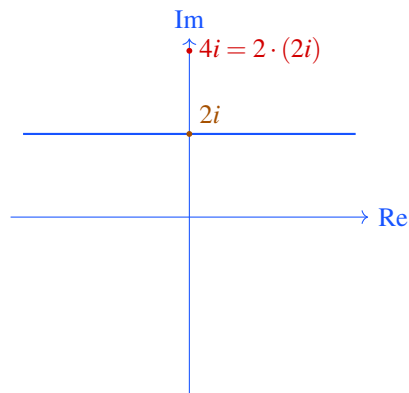
The points in orange are for $b = \pm 2$. Just like S , this is a line that *passes through the origin*, so it satisfies all the subspaces tests. We can find a basis by writing it as

$$\left\{ b \begin{bmatrix} 0 \\ 1 \end{bmatrix} \mid b \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\},$$

so a valid basis is $\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$.

(d) $V = \{v \mid v = a + 2i, a \in \mathbb{R}\}$.

Solution: No, V is not a subspace. We can draw it like this:



This is a line that *does not* pass through the origin (it is the horizontal line at height 2), so 0 is not in the set. Further, adding two points on this line, or scaling any point on this line, generally is not on the line. For example, $2 \cdot (2i) = 4i$ (red) lies above the line but $2i$ (orange) lies on it.

2. Operations on Subspaces

Let $\mathbb{V}$ be a vector space with subspaces $\mathbb{S}$ and $\mathbb{T}$.

Hint: in this problem, you are asked to write some more complicated proofs than you've encountered so far in this class. Fear not! It is generally good proof technique to start by simply writing down all the definitions you know. For example, what can you say about two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{S}$ using the fact that $\mathbb{S}$ is a subspace? Then think about how can you use these things you know to be true to show the desired conclusion.

- (a) Consider the set of vectors $\mathbb{S} + \mathbb{T} := \{\mathbf{s} + \mathbf{t} \mid \mathbf{s} \in \mathbb{S}, \mathbf{t} \in \mathbb{T}\}$. Show that $\mathbb{S} + \mathbb{T}$ is a subspace of $\mathbb{V}$.

Solution: Since $\mathbf{0} \in \mathbb{S}$ and $\mathbf{0} \in \mathbb{T}$, then $\mathbf{0} = \mathbf{0} + \mathbf{0} \in \mathbb{S} + \mathbb{T}$.

Let $\mathbf{v} = \mathbf{s}_1 + \mathbf{t}_1$ and $\mathbf{w} = \mathbf{s}_2 + \mathbf{t}_2$ be in $\mathbb{S} + \mathbb{T}$. $\mathbf{v} + \mathbf{w} = (\mathbf{s}_1 + \mathbf{s}_2) + (\mathbf{t}_1 + \mathbf{t}_2)$. Since $\mathbf{s}_1 + \mathbf{s}_2$ is a sum of vectors in $\mathbb{S}$, $\mathbf{s}_1 + \mathbf{s}_2 \in \mathbb{S}$. Likewise, $\mathbf{t}_1 + \mathbf{t}_2 \in \mathbb{T}$. Since $\mathbf{v} + \mathbf{w}$ is a sum of vectors in $\mathbb{S}$ and $\mathbb{T}$, $\mathbf{v} + \mathbf{w} \in \mathbb{S} + \mathbb{T}$, so $\mathbb{S} + \mathbb{T}$ is closed under vector addition. Consider a scalar, α . $\alpha\mathbf{v} = \alpha\mathbf{s}_1 + \alpha\mathbf{t}_1$. Since $\alpha\mathbf{s}_1 \in \mathbb{S}$ and $\alpha\mathbf{t}_1 \in \mathbb{T}$, $\alpha\mathbf{v} \in \mathbb{S} + \mathbb{T}$. $\mathbb{S} + \mathbb{T}$ is closed under scalar multiplication. Thus, $\mathbb{S} + \mathbb{T}$ is a subspace of $\mathbb{V}$.

- (b) Consider the intersection of $\mathbb{S}$ and $\mathbb{T}$: $\mathbb{S} \cap \mathbb{T} = \{\mathbf{v} \in \mathbb{V} \mid \mathbf{v} \in \mathbb{S}, \mathbf{v} \in \mathbb{T}\}$. Show that this is a subspace.

Hint: if $\mathbf{v} \in \mathbb{S} \cap \mathbb{T}$, this means $\mathbf{v} \in \mathbb{S}$ AND $\mathbf{v} \in \mathbb{T}$

Solution: Both subspaces $\mathbb{S}$ and $\mathbb{T}$ must contain $\mathbf{0}$ as they are subspaces. So $\mathbb{S} \cap \mathbb{T}$ also contains $\mathbf{0}$.

Let $\mathbf{v}, \mathbf{w} \in \mathbb{S} \cap \mathbb{T}$. So $\mathbf{v} \in \mathbb{S}$ and $\mathbf{w} \in \mathbb{S}$. Since $\mathbb{S}$ is a subspace, $\mathbf{v} + \mathbf{w} \in \mathbb{S}$. Likewise, we can say that $\mathbf{v} + \mathbf{w} \in \mathbb{T}$. Since $\mathbf{v} + \mathbf{w}$ is in both $\mathbb{S}$ and $\mathbb{T}$, $\mathbb{S} \cap \mathbb{T}$ is closed under vector addition.

Consider a scalar α . $\mathbf{v} \in \mathbb{S}$ means that $\alpha\mathbf{v} \in \mathbb{S}$ as $\mathbb{S}$ is closed under scalar multiplication. It is also the case that $\mathbf{v} \in \mathbb{T}$ and therefore $\alpha\mathbf{v} \in \mathbb{T}$. Since $\alpha\mathbf{v}$ is in both, $\alpha\mathbf{v} \in \mathbb{S} \cap \mathbb{T}$. $\mathbb{S} \cap \mathbb{T}$ is closed under scalar multiplication.

So $\mathbb{S} \cap \mathbb{T}$ is a subspace.

- (c) As before, $\mathbb{S}$ and $\mathbb{T}$ are subspaces of the vector space $\mathbb{V}$. Now, assume $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$. Let $B_S = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k\}$ be a basis for $\mathbb{S}$, and $B_T = \{\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_n\}$ be a basis for $\mathbb{T}$.

- i. Show that $B_S \cup B_T = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_k, \mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_n\}$ is a basis for $\mathbb{S} + \mathbb{T}$.

Hint 1: To show that these vectors span $\mathbb{S} + \mathbb{T}$, think about how you can write $\mathbf{s} \in \mathbb{S}$ in terms of the basis vectors in B_S ? A similar thing can be done for any $\mathbf{t} \in \mathbb{T}$.

Hint 2: To show linear independence, note that since $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$, if $\mathbf{s} \neq \mathbf{0}$ and $\mathbf{t} \neq \mathbf{0}$, then $\mathbf{s} + \mathbf{t} \neq \mathbf{0}$.

Solution: Since B_S and B_T are bases for $\mathbb{S}$ and $\mathbb{T}$, vectors $\mathbf{s} \in \mathbb{S}$ can be written as the linear combination $\mathbf{s} = \sum_{i=1}^k \alpha_i \mathbf{s}_i$ and vectors $\mathbf{t} \in \mathbb{T}$ can be written as the linear combination $\mathbf{t} = \sum_{j=1}^n \beta_j \mathbf{t}_j$. This means that vectors $\mathbf{v} \in \mathbb{S} + \mathbb{T}$ can be written as $\mathbf{s} + \mathbf{t} = \sum_{i=1}^k \alpha_i \mathbf{s}_i + \sum_{j=1}^n \beta_j \mathbf{t}_j$, so $\mathbb{S} + \mathbb{T}$ is spanned by $B_S \cup B_T$. Now it should suffice to show that the set of vectors $B_S \cup B_T$ is linearly independent. Since the intersection only contains $\mathbf{0}$, for any $\mathbf{s} \neq \mathbf{0}$ and $\mathbf{t} \neq \mathbf{0}$, $\mathbf{s} \neq \mathbf{t}$, $\mathbf{s} + \mathbf{t} \neq \mathbf{0}$. If this is the case, $\sum_{i=1}^k \alpha_i \mathbf{s}_i + \sum_{j=1}^n \beta_j \mathbf{t}_j \neq \mathbf{0}$ for any non-zero choices of α_i and β_j . This means the vectors in $B_S \cup B_T$ are linearly independent, thus we have a basis.

- ii. What is $\dim(\mathbb{S} + \mathbb{T})$ in terms of $\dim(\mathbb{S})$ and $\dim(\mathbb{T})$?

Solution: In the last part, we showed that the union of the bases of $\mathbb{S}$ and $\mathbb{T}$ is a basis for $\mathbb{S} + \mathbb{T}$ if $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$. Since none of $\mathbf{s}_i$ are in $\mathbb{T}$ and none of the $\mathbf{t}_j$ are in $\mathbb{S}$, the basis for $\mathbb{S} + \mathbb{T}$ will contain all basis vectors. However, each basis contains $\dim(\mathbb{S})$ and $\dim(\mathbb{T})$ vectors respectively. So in the basis for $\mathbb{S} + \mathbb{T}$, we have $\dim(\mathbb{S}) + \dim(\mathbb{T})$ vectors. By definition the dimension is the count of basis vectors $\dim(\mathbb{S} + \mathbb{T}) = \dim(\mathbb{S}) + \dim(\mathbb{T})$.

- (d) **(Optional)** For arbitrary subspaces $\mathbb{S}$ and $\mathbb{T}$ of vector space $\mathbb{V}$, show that $\dim(\mathbb{S} + \mathbb{T}) = \dim(\mathbb{S}) + \dim(\mathbb{T}) - \dim(\mathbb{S} \cap \mathbb{T})$. Observe that If $\mathbb{S} \cap \mathbb{T} = \{\mathbf{0}\}$, then $\dim(\mathbb{S} \cap \mathbb{T}) = 0$ and we recover the result from part (c).

Solution: Consider first a basis $B_{\mathbb{S} \cap \mathbb{T}}$ for $\mathbb{S} \cap \mathbb{T}$. This basis may not span $\mathbb{S}$ or $\mathbb{T}$. Extend $B_{\mathbb{S} \cap \mathbb{T}}$ to a basis for $\mathbb{S}$, $B_{\mathbb{S}}$, by adding linearly independent vectors from $\mathbb{S}$. Also create a basis for $\mathbb{T}$, $B_{\mathbb{T}}$, by adding linearly independent vectors from $\mathbb{T}$ to $B_{\mathbb{S} \cap \mathbb{T}}$. If we combine both $B_{\mathbb{S}}$ and $B_{\mathbb{T}}$, we will have enough vectors to span both $\mathbb{S}$ and $\mathbb{T}$ and therefore $\mathbb{S} + \mathbb{T}$. However, we must not double count the vectors in $B_{\mathbb{S}}$ and $B_{\mathbb{T}}$ that came from $B_{\mathbb{S} \cap \mathbb{T}}$. So the total count of basis vectors is $\dim(\mathbb{S})$ for the vectors in $B_{\mathbb{S}}$, $\dim(\mathbb{T})$ for the vectors in $B_{\mathbb{T}}$, with a count of the vectors in $B_{\mathbb{S} \cap \mathbb{T}}$ removed (subtract $\dim(\mathbb{S} \cap \mathbb{T})$).

$$\dim(\mathbb{S} + \mathbb{T}) = \dim(\mathbb{S}) + \dim(\mathbb{T}) - \dim(\mathbb{S} \cap \mathbb{T}).$$

3. Mechanical Basis

(a) Consider the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

- Are $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ linearly independent? Explain why.
- Can $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ form a basis for the vector space $\mathbb{R}^4$? Provide a succinct, yet clear and convincing explanation.

Solution:

i. The vectors are linearly independent. We can see that $\mathbf{v}_1$ is the only vector, with a non-zero second entry, so it cannot be a linear combination of the other two vectors. Similarly, $\mathbf{v}_2$ is the only vector with a non-zero final entry so it cannot be a linear combination of the other two vectors. Since two of the three vectors are linearly independent from the rest of the set, all three vectors must be linearly independent from one another.

ii. No, there are only 3 vectors, but in order to span $\mathbb{R}^4$, you need at least 4 linearly independent vectors so that every vector in $\mathbb{R}^4$ space can be formed by a linear combination of the basis vectors.

(b) Consider the set of vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \mathbf{v}_5 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

i. Explain why the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ is or is not a basis for $\mathbb{R}^3$.

ii. Consider $\mathbf{x} = \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}$.

Express $\mathbf{x}$ as a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$. Is this linear combination unique? If so, explain. If not, give an alternative linear combination.

Solution:

i. The set of five vectors is NOT a basis, since there are 5 vectors for a 3 dimensional space and a basis is defined as the set of minimum number of vectors required to span the vector space. In other words, a basis must consist of linearly independent vectors but this set of vectors is linearly dependent.

ii. One possible linear combination is:

$$3\mathbf{v}_1 + 3\mathbf{v}_2 + \mathbf{v}_4 = 3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 3+0+2 \\ 3+3+0 \\ 0+3+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}$$

However, because our vectors are linearly dependent, this linear combination is not unique. A couple other possible linear combinations include:

$$3\mathbf{v}_2 + \mathbf{v}_3 + \mathbf{v}_4 = 3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0+3+2 \\ 3+3+0 \\ 3+0+2 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}$$

$$\mathbf{v}_4 + 3\mathbf{v}_5 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+3 \\ 0+6 \\ 2+3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 5 \end{bmatrix}$$

4. Projections

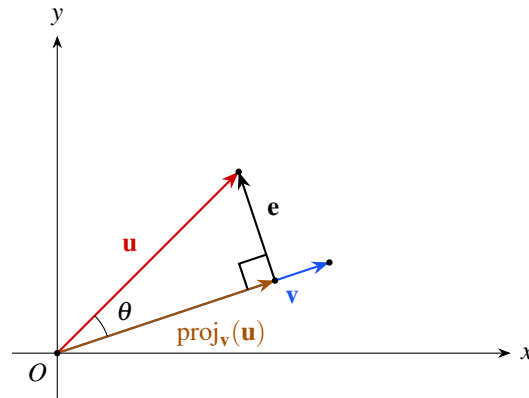


Figure 1: $\text{proj}_{\mathbf{v}}(\mathbf{u})$ visualized in $\mathbb{R}^2$

Figure 1 shows an example of an important concept in linear algebra known as **vector projection**, the act of "projecting" a vector onto another vector. This allows one to deconstruct any vector $\mathbf{u}$, given another vector $\mathbf{v}$, into two components:

- $\text{proj}_{\mathbf{v}}(\mathbf{u})$: a vector representing the component of $\mathbf{u}$ that points along another vector $\mathbf{v}$. This is also known as the **projection** of $\mathbf{u}$ onto $\mathbf{v}$.
- $\mathbf{e}$ is equal to $\mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u})$. In vector projection, $\mathbf{e}$ will be constructed such that $\|\mathbf{e}\|$ is minimized. Because of this, $\mathbf{e}$ will always be orthogonal to both $\mathbf{v}$ and $\text{proj}_{\mathbf{v}}(\mathbf{u})$. Intuitively, can you understand why? We will formally prove this later in the class.

In this problem, we will geometrically derive the formula for $\text{proj}_{\mathbf{v}}(\mathbf{u})$, and motivate the intuition behind vector projection.

- (a) Using Figure 1, calculate $\|\text{proj}_{\mathbf{v}}(\mathbf{u})\|$. You may use θ , $\|\mathbf{u}\|$, and $\|\mathbf{v}\|$ in your answer.

Solution:

Using the triangle in Figure 1, we can see that $\text{proj}_{\mathbf{v}}(\mathbf{u})$, $\mathbf{e}$, and $\mathbf{u}$ form a right triangle, with side lengths corresponding to their norms. Therefore, we can use trigonometry to see that $\|\text{proj}_{\mathbf{v}}(\mathbf{u})\| = \|\mathbf{u}\| \cos(\theta)$.

- (b) Now, using your answer to part (a), express $\text{proj}_{\mathbf{v}}(\mathbf{u})$ in terms of θ , $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, $\mathbf{u}$, and $\mathbf{v}$.

Solution:

From part (a), we know the magnitude of the vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$. Now, we must derive its direction, and ensure it does not change the magnitude we have already solved for. To do this, we must take a vector along our given direction, and normalize it.

Since we know that a projection onto $\mathbf{v}$ points along $\mathbf{v}$, then our normalized direction vector (typically known as a unit vector) will be $\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$. Now, we can see that our projection vector is given by: $\text{proj}_{\mathbf{v}}(\mathbf{u}) = \|\mathbf{u}\| \cos(\theta) \frac{\mathbf{v}}{\|\mathbf{v}\|}$.

- (c) Finally, express $\text{proj}_{\mathbf{v}}(\mathbf{u})$ in terms of $\mathbf{u}$, $\mathbf{v}$, and relevant inner products. You may not use norms or θ in your expression.

Hint: How can we rewrite terms that include norms and θ in terms of inner products?

Solution:

Remember that $\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$. In the previous subpart, we showed that $\text{proj}_{\mathbf{v}}(\mathbf{u}) = \|\mathbf{u}\| \cos(\theta) \frac{\mathbf{v}}{\|\mathbf{v}\|}$. Using our definition of $\langle \mathbf{u}, \mathbf{v} \rangle$, we can see that:

$$\|\mathbf{u}\| \cos(\theta) \frac{\mathbf{v}}{\|\mathbf{v}\|} = \|\mathbf{u}\| \|\mathbf{v}\| \cos(\theta) \frac{\mathbf{v}}{\|\mathbf{v}\|^2} = \langle \mathbf{u}, \mathbf{v} \rangle \frac{\mathbf{v}}{\|\mathbf{v}\|^2}$$

Now, let's try and get rid of the $\|\mathbf{v}\|^2$ term. We can do this using $\langle \mathbf{v}, \mathbf{v} \rangle$! Since $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$:

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle \frac{\mathbf{v}}{\|\mathbf{v}\|^2} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle} \mathbf{v}$$

This is the standard formula for vector projection!

5. Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality states that for two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$|\langle \mathbf{v}, \mathbf{w} \rangle| = |\mathbf{v}^\top \mathbf{w}| \leq \|\mathbf{v}\| \cdot \|\mathbf{w}\|$$

In this problem we will prove the Cauchy-Schwarz inequality for vectors in $\mathbb{R}^2$.

Take two vectors: $\mathbf{v} = r \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$ and $\mathbf{w} = t \begin{bmatrix} \cos \phi \\ \sin \phi \end{bmatrix}$, where $r > 0$, $t > 0$, $\theta \in [0, 2\pi]$, and $\phi \in [0, 2\pi]$ are scalars. Make sure you understand why any vector in $\mathbb{R}^2$ can be expressed this way and why it is acceptable to restrict $r, t > 0$.

- (a) In terms of some or all of the variables r, t, θ , and ϕ , what are $\|\mathbf{v}\|$ and $\|\mathbf{w}\|$?

Hint: Recall the trigonometric identity: $\cos^2 x + \sin^2 x = 1$.

Solution: We use the trigonometric identity $\cos^2 x + \sin^2 x = 1$ to show:

$$\begin{aligned} \|\mathbf{v}\| &= \sqrt{v_1^2 + v_2^2} \\ &= \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \\ &= r \sqrt{\cos^2 \theta + \sin^2 \theta} \\ &= r \end{aligned}$$

Similarly, $\|\mathbf{w}\| = t$.

- (b) In terms of some or all of the variables r, t, θ , and ϕ , what is $\langle \mathbf{v}, \mathbf{w} \rangle$?

Hint: The trigonometric identity $\cos a \cos b + \sin a \sin b = \cos(a - b)$ may be useful.

Solution: We use the trigonometric identity $\cos(x) \cos(y) + \sin(x) \sin(y) = \cos(x - y)$ to show:

$$\begin{aligned} \langle \mathbf{v}, \mathbf{w} \rangle &= (r \cos \theta)(t \cos \phi) + (r \sin \theta)(t \sin \phi) \\ &= r \cdot t (\cos \theta \cos \phi + \sin \theta \sin \phi) \\ &= r \cdot t \cos(\theta - \phi). \end{aligned}$$

Note that this formula matches our intuition of inner products as measuring how close the directions of two vectors are since it includes since the last term directly depends on the difference in the angles of the two vectors. It is maximized when the angle difference is zero!

- (c) Now we will show that the Cauchy-Schwarz inequality holds for any two vectors in $\mathbb{R}^2$. To do this we must show that $\langle \mathbf{v}, \mathbf{w} \rangle$ is upper-bounded by $\|\mathbf{v}\| \|\mathbf{w}\|$ and lower-bounded by $-\|\mathbf{v}\| \|\mathbf{w}\|$. For this part, show that $\langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\| \|\mathbf{w}\|$.

Hint: consider your results from part (b). Also recall $\cos x \leq 1$

Solution: We use the fact that $\cos x \leq 1$ to show:

$$\begin{aligned} \langle \mathbf{v}, \mathbf{w} \rangle &= r \cdot t \cos(\theta - \phi) \\ &= \|\mathbf{v}\| \|\mathbf{w}\| \cos(\theta - \phi) \\ &\leq \|\mathbf{v}\| \|\mathbf{w}\| \end{aligned}$$

- (d) Now show that $\langle \mathbf{v}, \mathbf{w} \rangle \geq -\|\mathbf{v}\| \|\mathbf{w}\|$.

Hint: consider your results from part (b). Also recall $\cos x \geq -1$.

Solution:

We use the fact that $\cos x \geq -1$ to show:

$$\begin{aligned}\langle \mathbf{v}, \mathbf{w} \rangle &= r \cdot t \cos(\theta - \phi) \\ &= \|\mathbf{v}\| \|\mathbf{w}\| \cos(\theta - \phi) \\ &\geq -\|\mathbf{v}\| \|\mathbf{w}\|\end{aligned}$$

Therefore, in total:

$$-\|\mathbf{v}\| \|\mathbf{w}\| \leq \langle \mathbf{v}, \mathbf{w} \rangle \leq \|\mathbf{v}\| \|\mathbf{w}\|,$$

which gives us that

$$|\langle \mathbf{v}, \mathbf{w} \rangle| \leq \|\mathbf{v}\| \|\mathbf{w}\|.$$

- (e) Note that the inequality states that the absolute value of the inner product of two vectors must be less than *or equal to* the product of their magnitudes. What conditions must the vectors satisfy for the equality to hold? In other words, when is $|\langle \mathbf{v}, \mathbf{w} \rangle| = \|\mathbf{v}\| \cdot \|\mathbf{w}\|$?

Solution:

$$\begin{aligned}|\langle \mathbf{v}, \mathbf{w} \rangle| &= \|\mathbf{v}\| \|\mathbf{w}\| \\ \|\mathbf{v}\| \|\mathbf{w}\| \cos(\theta - \phi) &= \|\mathbf{v}\| \|\mathbf{w}\| \\ |\cos(\theta - \phi)| &= 1\end{aligned}$$

If $\cos(\theta - \phi) = 1$, then $\theta = \phi$. Otherwise if $\cos(\theta - \phi) = -1$, then $\theta - \phi = \pi$. We see that the equality holds when the angle between the two vectors is 0 or π . Note that when the angle is 0 or π , the vectors would be linearly dependent.

6. Triangular Reasoning

- (a) Let the vectors $\begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ serve as the legs of a triangle. What is the length of the hypotenuse of this triangle?

Solution:

Besides the traditional use of the Pythagorean Theorem, we can also calculate the hypotenuse length as the distance between our leg vectors.

$$\mathbf{v} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

$$\|\mathbf{v}\| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

The hypotenuse is of length 5. Was our process actually that different from the usual geometric approach?

- (b) Let us define the following two vectors:

$$\mathbf{x} = \begin{bmatrix} -3 \\ 3 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

What is the distance between these two vectors?

Hint: It may be useful to recognize that $\|\mathbf{x}\| = 3\sqrt{2}$ and $\|\mathbf{y}\| = 4\sqrt{2}$. What is the angle between the two vectors?

Solution:

We could solve for the distance by explicitly using the process we have learned. However, observing the similarities between these vectors and the triangle from last problem can save us some time. In particular we can see that $\mathbf{x}$ and $\mathbf{y}$ act as legs in a right triangle, being rotated 90 degrees from one another. Indeed they are actually the same legs as the triangle from part (a), just scaled up by $\sqrt{2}$. As a result, their distance, the hypotenuse of their triangle, will simply be $\sqrt{2}$ times the hypotenuse from part (a). Therefore, the vectors are a distance of $5\sqrt{2} = \sqrt{50}$ apart.

- (c) Use the Cauchy-Schwarz inequality to verify (i.e. prove or derive) the triangle inequality:

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

Solution: We consider the Euclidean 2-norm here. By Cauchy-Schwarz inequality we have

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

$$|\langle \mathbf{y}, \mathbf{x} \rangle| \leq \|\mathbf{y}\| \cdot \|\mathbf{x}\|$$

Therefore,

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y})^\top (\mathbf{x} + \mathbf{y}) \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle \\ &\leq \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2(\|\mathbf{y}\| \cdot \|\mathbf{x}\|) \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \end{aligned}$$

Taking square root on both sides, we get

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$$

7. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*

Solution:

I worked alone because my study group was out of town.

I prefer to work alone because a group can be very distracting for me.

I worked with my friends Alice and Bob. It's nice to be able to bounce ideas off one another and fill in each other's gaps. Explaining my ideas to someone else is hard, but it really makes sure I understand them well!

I met some new friends Carol and Danny at homework party, and we were able to finish most of the homework together with occasional help from a tutor. I reviewed my answers on my own to make sure I understood everything.

- (b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.

Solution:

I didn't remember some of the definitions I learned in linear algebra, so I consulted my old notes and reviewed concepts as I needed to while solving the problems

I went to office hours to work on the homework, and whenever I got stuck, I asked one of the tutors to provide some hints or review concepts I didn't remember.

I tried all the problems on my own, and then asked an LLM to critique my approach. I got 90% of the way there on my own, but this interaction enriched my understanding of the material!

I reviewed my lecture notes before starting the homework; and whenever I got stuck, I scoured the discussion worksheets for similar problems and used the solutions to guide my thought process.

I asked some questions on Ed whenever I was confused so that I could get the most fitting answer to my question, in the context of EE66. I also answered some questions because this is a good way to practice the stuff I know well.

I couldn't even understand what problem 2 was asking, so I asked an AI tool to review all the relevant prerequisite knowledge and explain the problem statement in a different way before trying to tackle it on my own.

- (c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?

Solution:

I found question 3 very intriguing because this is a totally different perspective on DTFS to what we learned in lecture. Question 2 was pretty boring and mechanical, because we were just practicing the synthesis equation, but I suppose it is important to be able to do these more-basic things too.

- (d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able practice in this assignment?

Solution: I did feel prepared! But, question 5 seemed to me to be out of place because it's been two weeks since we covered least squares. We talked about SVD and its application for PCA in class and discussion this week, and I didn't really understand any of it, so it would have been nice to have some problems which gave me an opportunity to practice those concepts.

(e) Is there anything else you would like to share?

Solution:

Johnny, who is a tutor, was SUPER DUPER helpful in office hours. He explained stuff in such a way that I still felt like I was discovering the answer to the problems by myself, which was awesome.

I'm feeling very overwhelmed by this class and the upcoming midterm is really stressing me out. These homeworks are really hard and I wish there were more resources available, which I could use to understand them deeply.